Segment-04

Signal and waveforms

Signal : Signals are variables that carry information.

Example : Audio, Vedio, Electrical signals (Voltage & current)

System : System is an assemblage of entity/objects, real/abstract comprising a whole with each every component interecting with one another.

→ Process input signal to find output signal.

Input Signal → $\boxed{\text{System}}$ → Output Signal

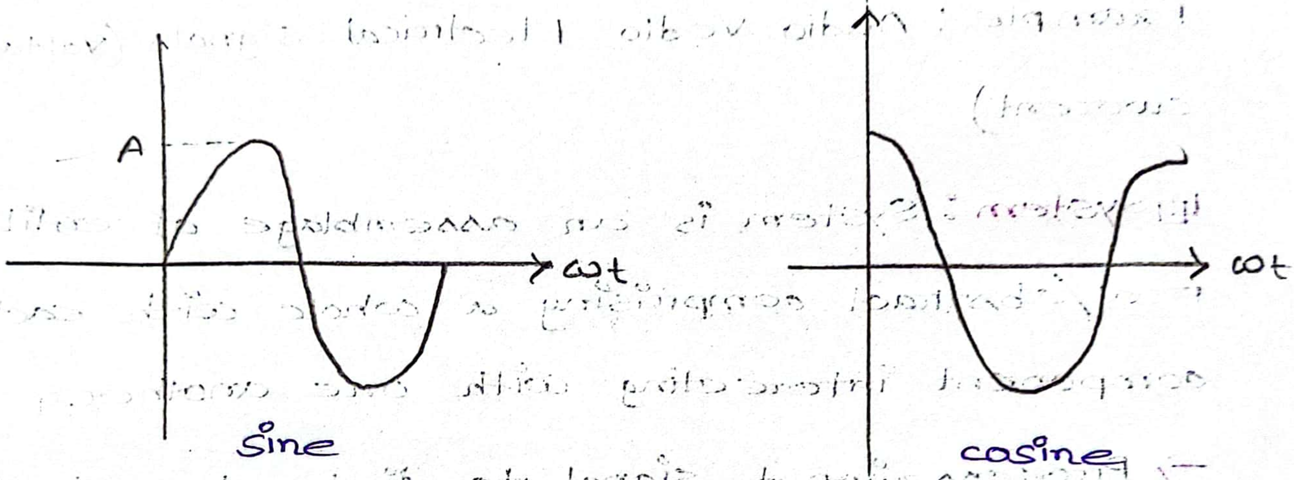
Example : Natural system

Human made system (computer programs, processor, machines, computer storge system etc)

Waveform: Shape of the signal is called waveform.

→ Sinusidal (sine, cosine)

→ Non-sinusidal

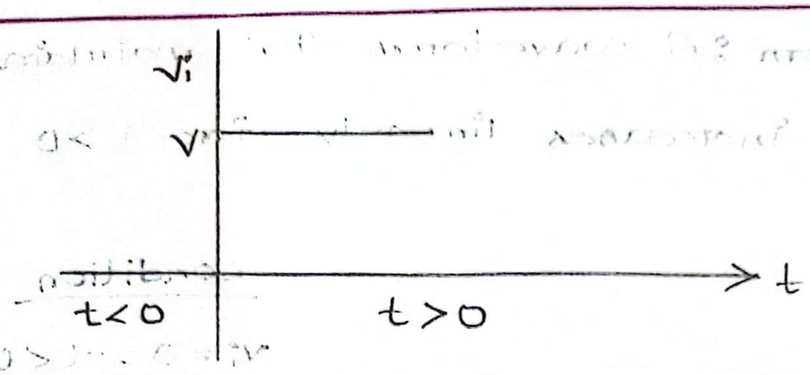


A = Amplitude of the waveform
(Maximum value of the)

- Time period, $T = \frac{1}{f}$

*** Non-Sinusidal :-**

1. **Step waveform**: A waveform that maintains zero value for time $t<0$ and maintains V for all the time $t>0$.

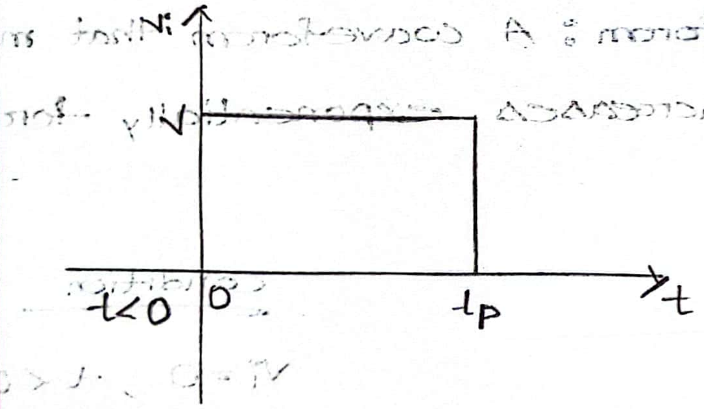


<u>condition</u>

$V_i = 0, \; t < 0$

$V_i = V, \; t > 0$

2\. **Pulse waveform :** A waveform that maintains 0 for $t < 0$ and maintains V for time $t = 0$ to t_p, where t_p is the pulse duration.

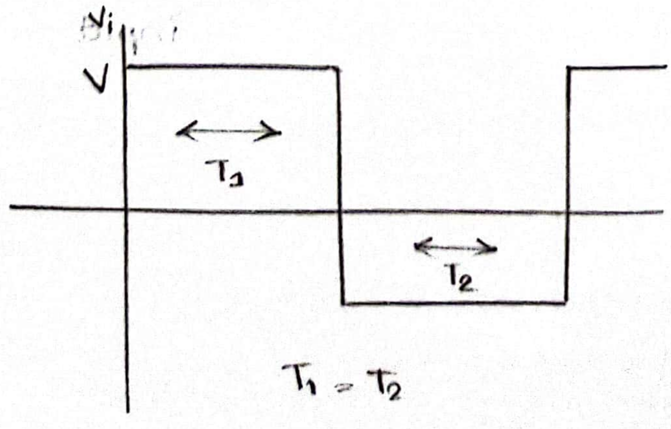


<u>condition</u>

$V_i = 0, \; t < 0$

$V_i = V, \; 0 \leq t \leq t_p$

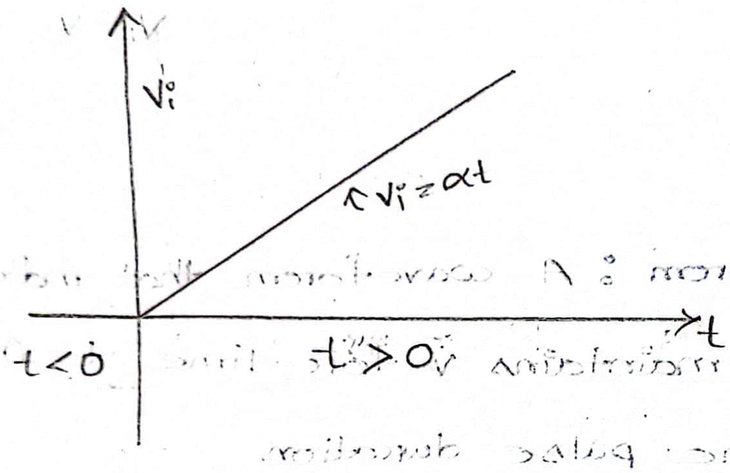
3\. **Square waveform :** A waveform that maintains one constant level for a time T_1 and another constant level for a time T_2.



<u>Condition</u>

$T = T_1 + T_2$

4. Ramp Waveform : A waveform that maintains 0 for $t<0$ and increases linearly for $t>0$

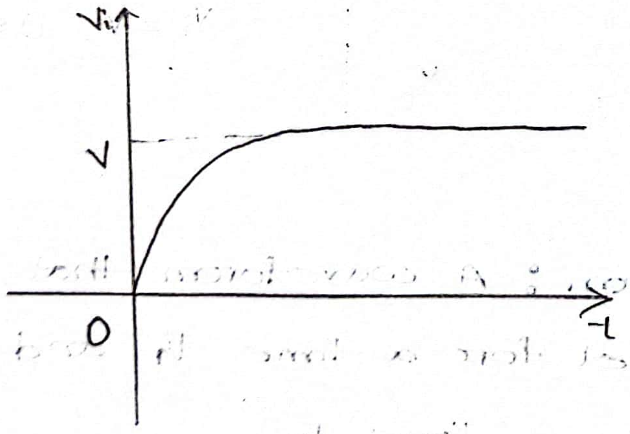


condition

$V_i = 0, \; t<0$

$V_i = \alpha t, \; t>0$

5. Exponential waveform : A waveform that maintains 0 for $t<0$ and increases exponentially for $t>0$.



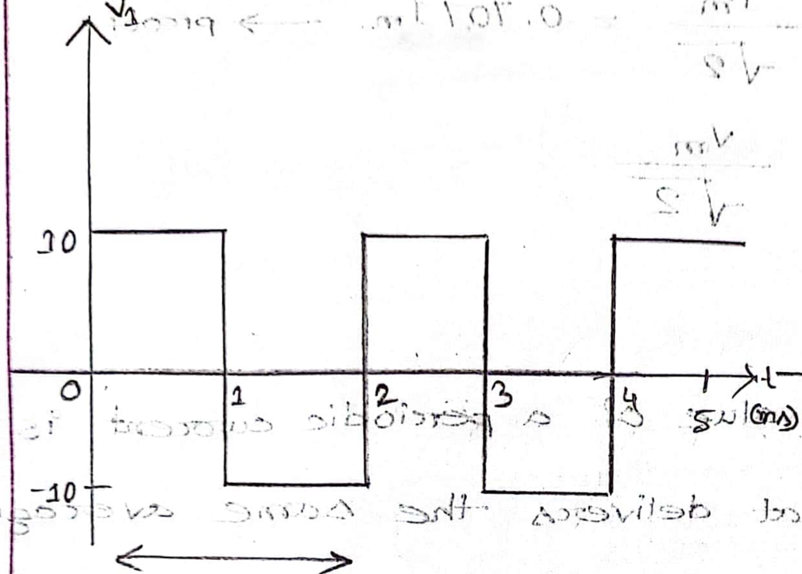
condition

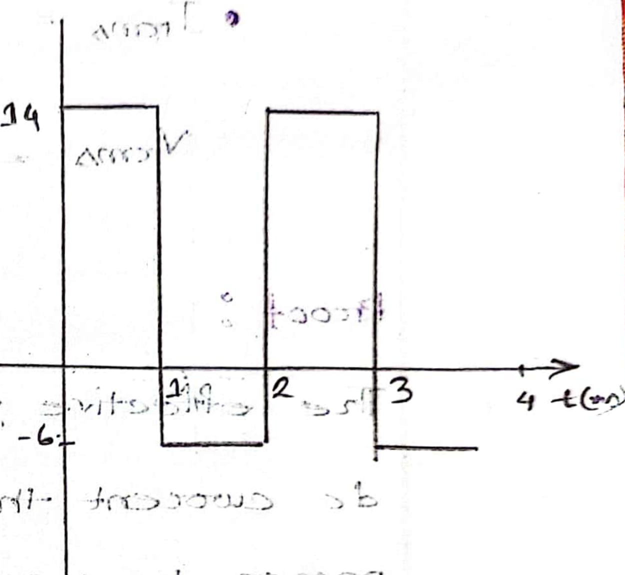
$V_i = 0 \; , \; t<0$

$V_i = V\left(1 - e^{-t/\tau}\right) t>0$

τ = Time constant of exponential input

Determine the average value of the waveforms:



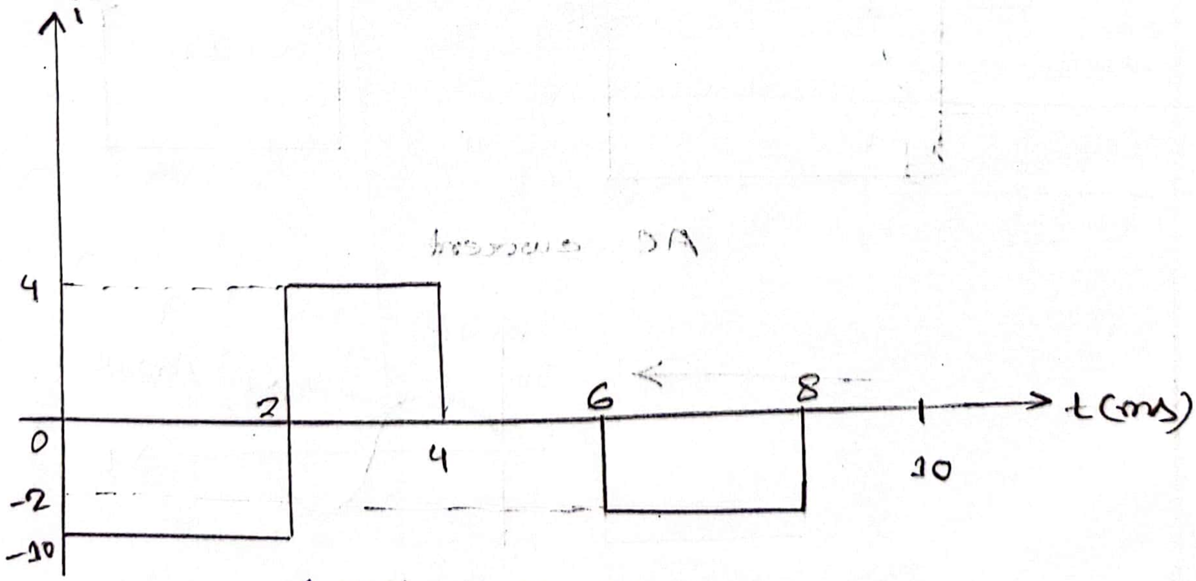


$$V_1(avg) = \frac{(\overset{\text{Amplitude}}{10} \times \overset{\text{Time duration}}{1}) + (-10 \times 1)}{2}$$

$$= 0V$$

$$V_2(avg) = \frac{(14 \times 1) + (-6 \times 1)}{2}$$

$$= 4V$$



1 cycle = T = 10 ms

$$i(avg) = \frac{(-10 \times 2) + (4 \times 2) + (-2 \times 2)}{10} = -1.6V$$

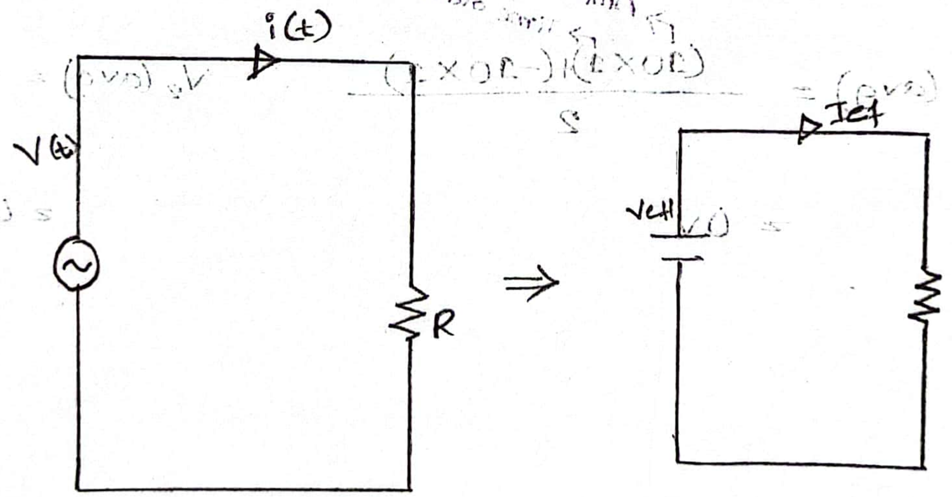
Effective or RMS value :

- $I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707\, I_m \longrightarrow$ proof

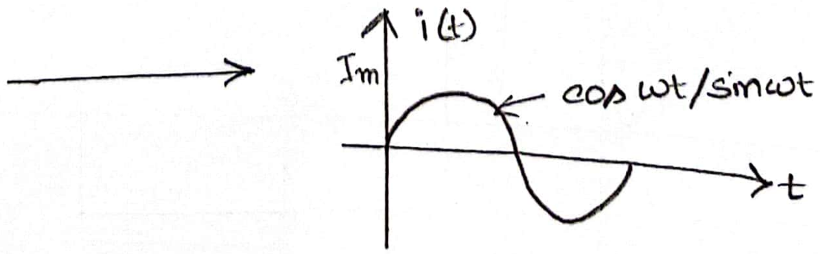
$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

Proof :

The effective value of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.



AC current



For a sinusoid,

$$i(t) = I_m \cos\omega t$$

$$I_{(rms)} = \sqrt{\frac{1}{T}\int_0^T i^2(t)\,dt}$$

$$= \sqrt{\frac{1}{T}\int_0^T I_m^2 \cos^2\omega t\,dt}$$

$$= \sqrt{\frac{I_m^2}{2T}\int_0^T 2\cos^2\omega t\,dt}$$

$$= \sqrt{\frac{I_m^2}{2T}\int_0^T (1+\cos 2\omega t)\,dt}$$

$$= \sqrt{\frac{I_m^2}{2T}\left[t\right]_0^T + \frac{I_m^2}{2T}\cdot\left[\frac{\sin 2\omega t}{2\omega}\right]_0^T}$$

$$= \sqrt{\frac{I_m^2}{2T}(T-0) + \frac{I_m^2}{2T}\left[\frac{\sin 2\cdot\frac{2\pi}{\cancel{T}}\cdot\cancel{T}}{2\cdot\frac{2\pi}{T}} - \frac{\sin 2\cdot\frac{2\pi}{T}\cdot 0}{2\cdot\frac{2\pi}{T}}\right]}$$

$$= \sqrt{\frac{I_m^2}{2}}$$

$$\therefore I_{rms} = \frac{I_m^2}{\sqrt{2}}$$

Proved

Same as V_{rms}

Segment - 5

Sinusoids and phasors

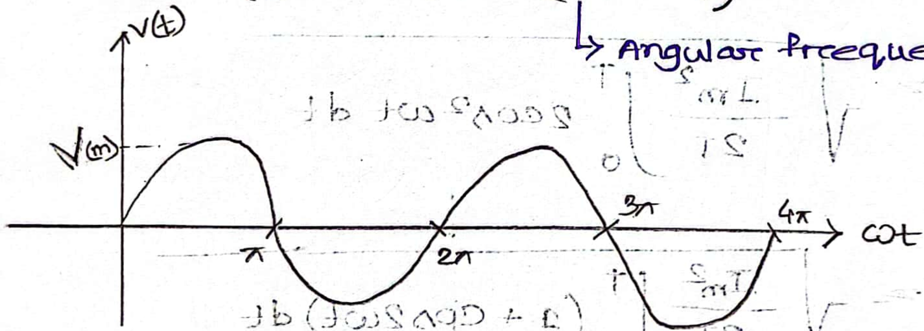
Sinusoid : A sinusoid is a signal that has the form of sine or cosine function.

→ General expression for sinusoid :

$$v(t) = V_m \sin(\omega t + \psi)$$

ψ → phase angle

ω → Angular frequency



Q. * Given a sinusoid, $5\sin(4\pi t - 60°)$

Calculate :

amplitude = 5

phase = $-60°$

angular frequency = 4π

period = $\frac{1}{2}$ s

frequency = 2 Hz

$f = \frac{4\pi}{2\pi}$

$= 2$

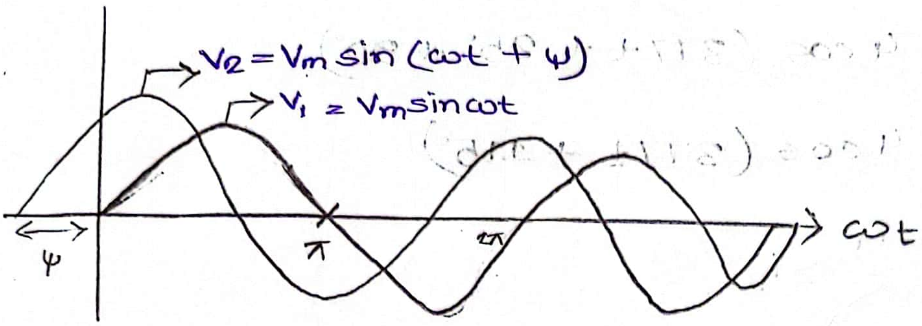
$\omega = 2\pi f$

$\therefore f = \frac{\omega}{2\pi}$

$T = \frac{1}{f}$

$= \frac{1}{2}$

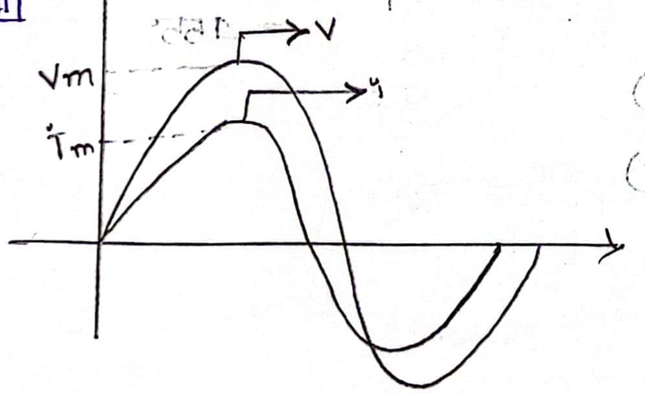
Phase of sinusoid :



⇒ V_2 leads V_1 by ψ ;

→ V_1 lags V_2 by ψ ;

* ⇒ V_1 and V_2 are out of phase



⇒ V and i are in-phase

Formula/Rules:

* $\sin(\omega t \pm 180^\circ) = -\sin\omega t$

* $\cos(\omega t \pm 180^\circ) = -\cos\omega t$

* $\sin(\omega t \pm 90^\circ) = \pm\cos\omega t$

* $\cos(\omega t \pm 90^\circ) = \mp\sin\omega t$

Q. * find the phase angle between

$i_1 = -4\sin(377t + 25^\circ)$

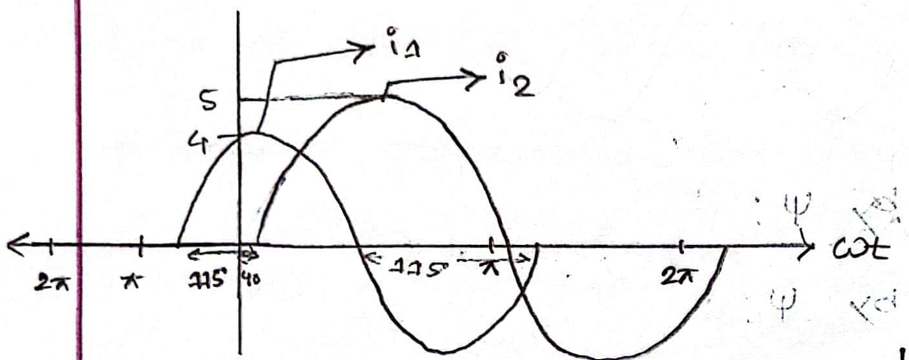
$i_2 = 5\cos(377t - 40^\circ)$

does i_1 lead or lag i_2 ?

Ans : $i_1 = -4\sin(377t + 25^\circ)$

$= 4\cos(377t + 25^\circ + 90^\circ)$

$= 4\cos(377t + 115^\circ)$



i_1 leads i_2 by 155°

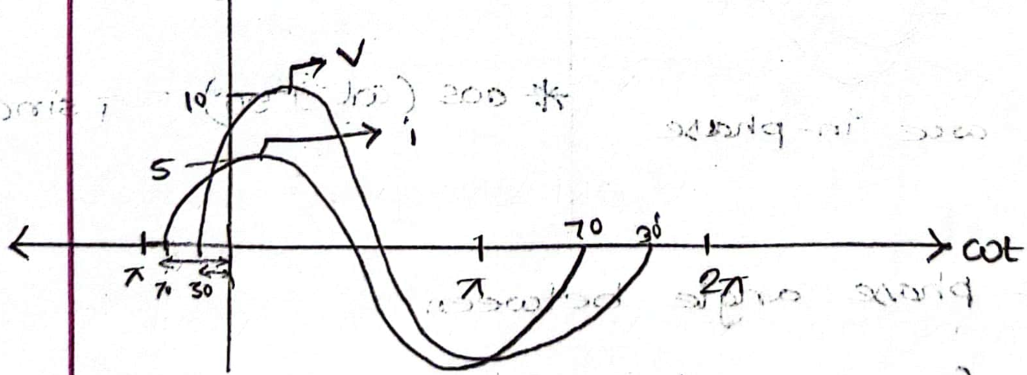
phase difference

$\psi = 115 - (-40)$

$= 155^\circ$

Q * $V = 10\sin(\omega t + 30^\circ)$

$i = 5\sin(\omega t + 70^\circ)$



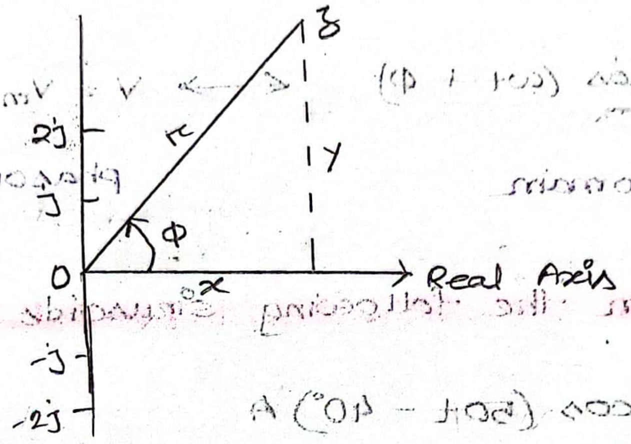
i leads V by $\psi = (70 - 30) = 40^\circ$

or,

V lags i by $\psi = 40^\circ$

Phasor (1)

- A phasor is a complex number that represents the amplitude and phase of a sinusoid.



a. Rectangular : $z = x + jy = r(\cos\phi + j\sin\phi)$

b. Polar : $z = r\angle\phi$

c. Exponential : $z = re^{j\phi}$

where, $r = \sqrt{x^2+y^2}$

$\phi = \tan^{-1}\dfrac{y}{x}$

Phasor (2)

a. $[(5+j2)(-1+j4) - 5\angle 60^\circ]$

$\dfrac{36-5\sqrt{3}}{2}$

using calculator →

Ans : $-15.5 + 13.67j$

b. $\dfrac{10+5j+3\angle 40^\circ}{-3+j4} + 10\angle 30^\circ$

Ans : $8.293 + 2.2j$.

Phasor (4)

- Transform a sinusoid to and from the time domain to the phasor domain:

$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow V = V_m \angle \phi$$

time domain　　　　phasor domain

Q * Transform the following sinusoids to phasor:

a. $i = 6\cos(50t - 40^\circ)$ A

b. $V = -4\sin(30t + 50^\circ)$ V

Ans: a. $I = 6\angle 40^\circ$ A

Ans: b. Since $-\sin A = \cos(A + 90^\circ)$:

$$v(t) = 4\cos(30t + 50^\circ + 90^\circ)$$

$$= 4\cos(30t + 140^\circ) \text{ V}$$

$$\therefore V = 4\angle 140^\circ \text{ V}$$

Q * Transform the sinusoids corresponding to phasors:

a. $V = -10\angle 30^\circ$ V

b. $I = j(5 - j12)$ A

Ans: a. $v(t) = 10\cos(\omega t + 210^\circ)$ V

Ans: b. Since,

$$I = 12 + j5 \, \text{A}$$

$$= \sqrt{12^2 + 5^2} \angle \tan^{-1}\left(\frac{5}{12}\right)$$

$$= 13\angle 22.62^\circ$$

$$\therefore i(t) = 13\cos(\omega t + 22.62^\circ) \text{ A}$$

Q * a. $I = -3 + j4$ A

b. $V = j8e^{-j20}$ V

Ans: a. $I = -3 + j4$ A

$$= \sqrt{(-3)^2 + (4)^2} \angle \tan^{-1}\left(-\frac{4}{3}\right) \rightarrow -53.10 + 180^\circ$$

$$= 5\angle 126.9^\circ$$

$$\therefore i(t) = 5\cos(\omega t + 126.9^\circ) \text{ A}$$

Ans: b. Since $j = 1\angle 90^\circ$

$$V = j8\angle -20$$

$$= (1\angle 90^\circ)(8\angle -20^\circ)$$

$$= 8(\angle 90^\circ - \angle 20^\circ) = 8\angle 70^\circ \text{ V}$$

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$\therefore v(t) = 8\cos(\omega t + 70^\circ)\,V$

Phasor (8)

* Relationship between differential, integral, operation in phasor listed as follow:

$$v(t) \longleftrightarrow V = V\angle\phi$$

$$\frac{dv}{dt} \longleftrightarrow j\omega V$$

$$\int v\,dt \longleftrightarrow \frac{V}{j\omega}$$

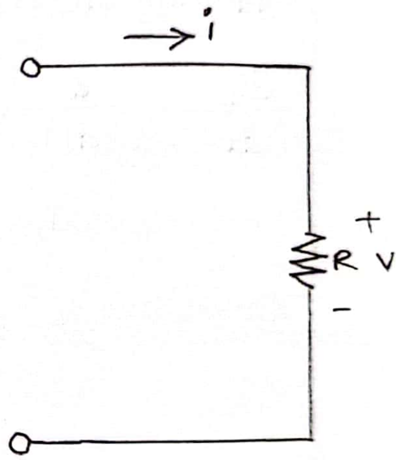
Q Problem * Use phasor approach, determine the ~~current~~ $i(t)$ in a circuit described by the integro-differential equation. (Ans: $i(t) = 4.642\cos(2t + 143.2^\circ)$ A

$$4i + 8\int i\,dt - 3\frac{di}{dt} = 50\cos(\underbrace{2t}_{\omega t} + 75^\circ)$$

Ans:

$$4I + 8.\frac{I}{j\omega} - 3.j\omega I = 50\angle 75^\circ$$

Phasor Relationships for circuit Elements (1)



• **Resistor:**

For $V = V_m \sin\omega t$

$$i = \frac{V}{R} = \frac{V_m \sin\omega t}{R}$$

$$= \frac{V_m}{R} \sin\omega t$$

$$= I_m \sin\omega t$$

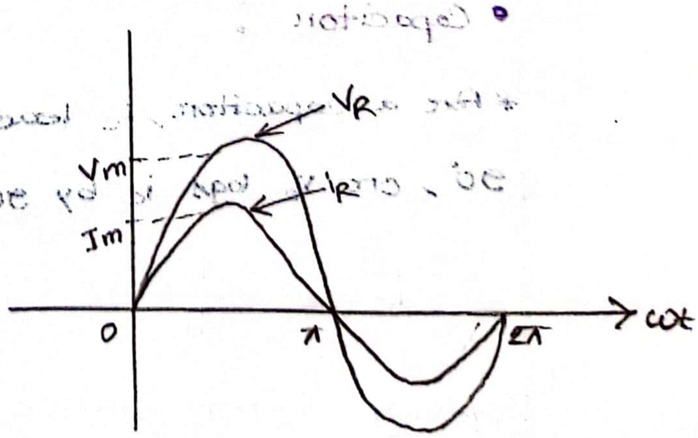
• in phase

• $I_m = \frac{V_m}{R}$

For a given i,

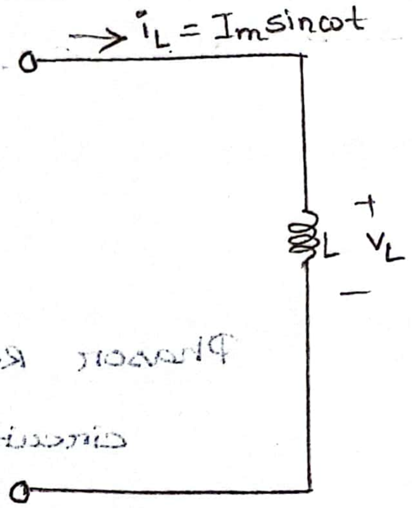
$$V = iR = (I_m \sin\omega t)R$$

$$= V_m \sin\omega t$$



• $V_m = I_m R$

• Inductor :



* An inductor V_L leads i_L by $90°$ or, i_L lags V_L by $90°$.

$$V_L = L \frac{di_L}{dt}$$

Applying differentiation,

$$\frac{di_L}{dt} = \frac{d}{dt}(I_m \sin\omega t)$$

$$= \omega I_m \cos\omega t$$

Therefore, $$V_L = L\frac{di_L}{dt}$$

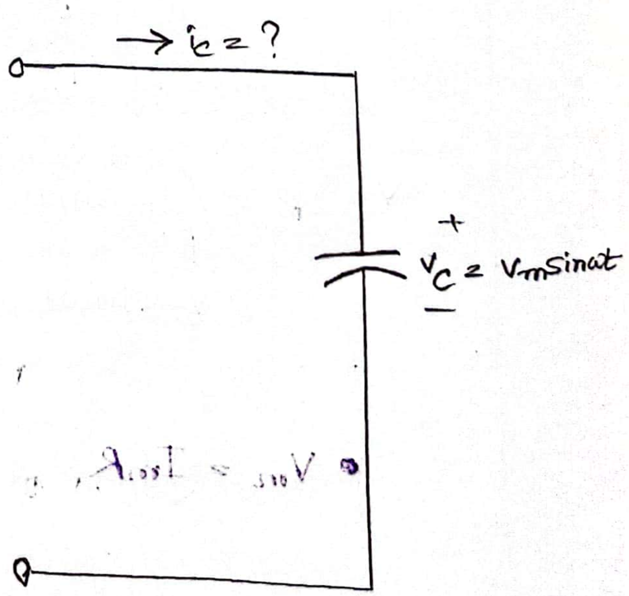
$$= L(\omega I_m \cos\omega t)$$

$$= \omega L I_m \cos\omega t$$

* $V_m = \omega L I_m$

or, $$V_L = V_m \sin(\omega t + 90°)$$

• Capacitor :



* For a capacitor, i_c leads V_c by $90°$, or V_c lags i_c by $90°$

$$i_c = C\frac{dV_c}{dt}$$

Applying differentiation,

$$\frac{dV_c}{dt} = \frac{d}{dt}(V_m \sin\omega t)$$

$$= \omega V_m \cos\omega t$$

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Therefore, $i_c = C\dfrac{dv_c}{dt}$

$$= C(\omega V_m \cos\omega t)$$

$$= \omega C V_m \cos\omega t$$

or, $i_c = I_m \sin(\omega t + 90^\circ)$

$$\therefore I_m = \omega C V_m$$

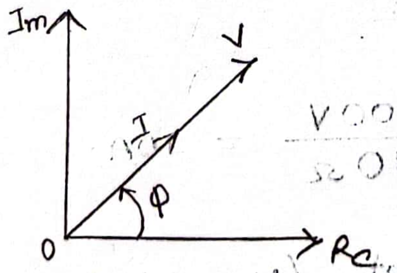
যে পরিমাণ বাধা দেয়

$$* \; X_C = \frac{V_m}{I_m} = \frac{V_m}{\omega C V_m} = \frac{1}{\omega C}$$

Capacitor এর ~~Capitancy~~ Capacitance or, reactance

- **Circuit :**

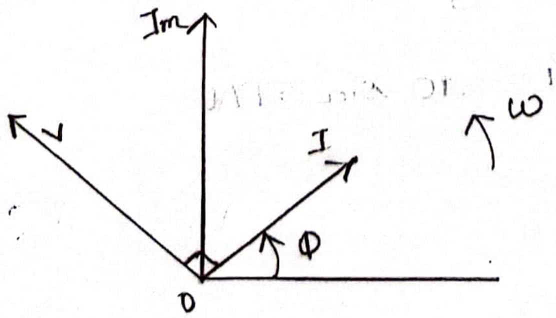
* Resistor এর জন্য phasor Diagram:



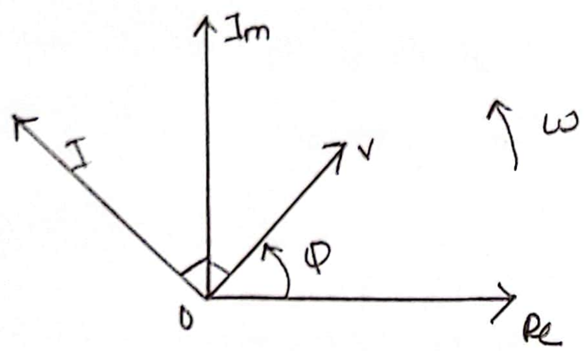
Voltage এবং current এর মধ্যে difference নাই, $\phi = 0$
In-phase
তাই যেখান থেকে I, সেখানেই V,

* Inductor এর জন্য phasor Diagram:

V, I কে 90° তে lead করে



* Capacitor এর জন্য phasor Diagram :



I, V কে lead করে,

Q * Example: 14.1 | The voltage across a resistor is indicated find the sinusoidal expression for the current if the resistor is 10Ω. Sketch the curves for v and i.

a. $V = 100 \sin 377t$

b. $V = 25 \sin(377t + 60°)$

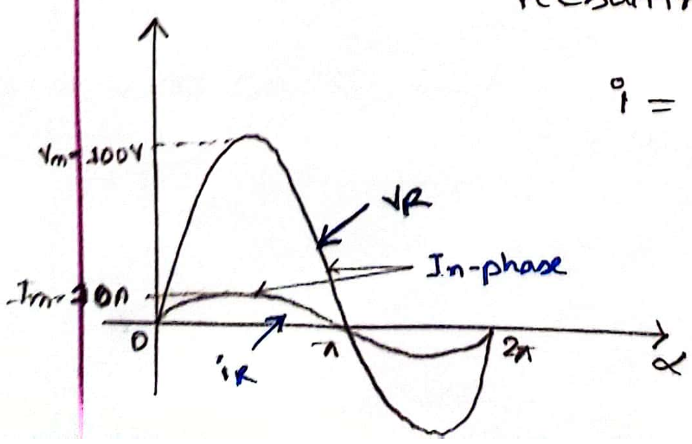
⇒ We know,

Ans: a

$$I_m = \frac{V_m}{R} = \frac{100V}{10\Omega} = 10A$$

It's a resistor, so that (v and i are in-phase) resulting in,

$$i = 10 \sin 377t$$

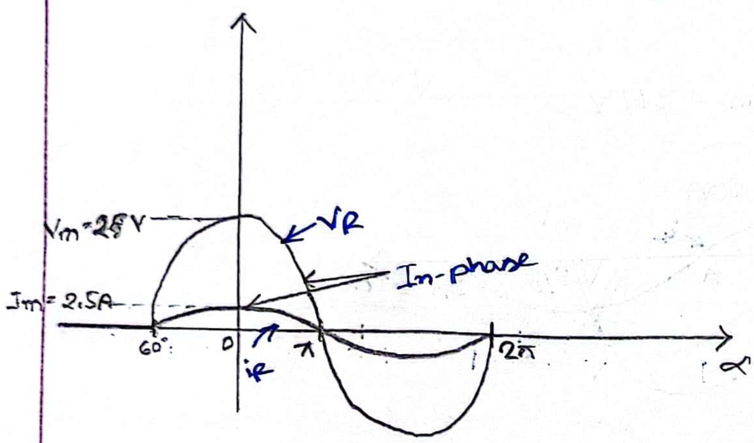


⇒ we know,

Ans: b. $I_m = \frac{V_m}{R} = \frac{25\,V}{10\,\Omega} = 2.5\,A$

It's a resistor, so that (v and i are in-phase)

resulting in, $i = 2.5 \sin(377t + 60°)$



Q * Example: 14.3 | The current through a 0.1-H ~~coil~~/inductor is provided. Find the sinusoidal expression for the voltage across the ~~coil~~ coil. Sketch the v and i curves.

a. $i = 10 \sin 377t$

b. $i = 7 \sin(377t - 70°)$

⇒ we know,

Ans: a. $X_L = \omega L$

$= (377\ rad/s)(0.1\,H)$

$= 37.7\,\Omega$

$V_m = I_m X_L = (10A)(37.7\,\Omega) = 377\,V.$

এখানে, unit হচ্ছে Ω

* X_L হচ্ছে রেজিস্টর এর জন্য ~~যে~~ Resistance, এখানে Inductor এর জন্য X_L হয়ে যাবে Inductance

* Ohm's Law, $V_m = I_m R$

↳ Inductor - এর জন্য. হবে,

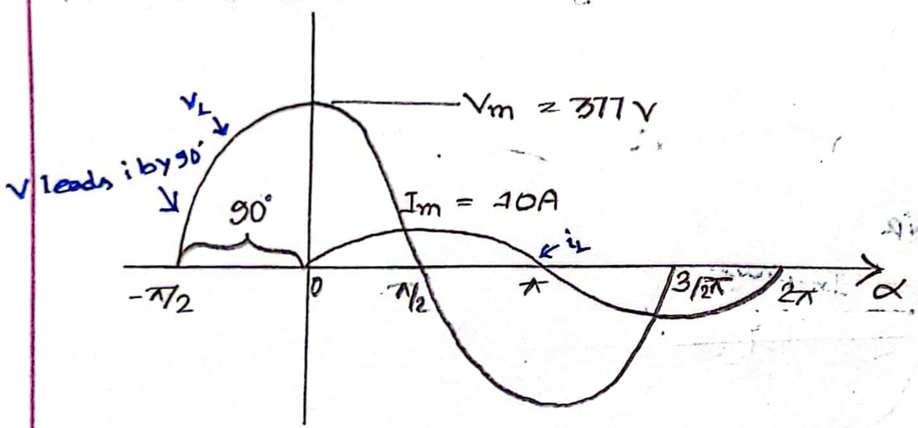
$V_m = I_m X_L$

Also we know that for a coil/inductor v leads i by

90°, therefore,

$$v = 377 \sin(377t + 90^\circ)$$

The curves are sketched,



⇒ from 'a' we got,

Ans: b. X_L remains at 37.7Ω

And, $V_m = I_m X_L$

$= (7A)(37.7\Omega)$

$= 263.9\,V$

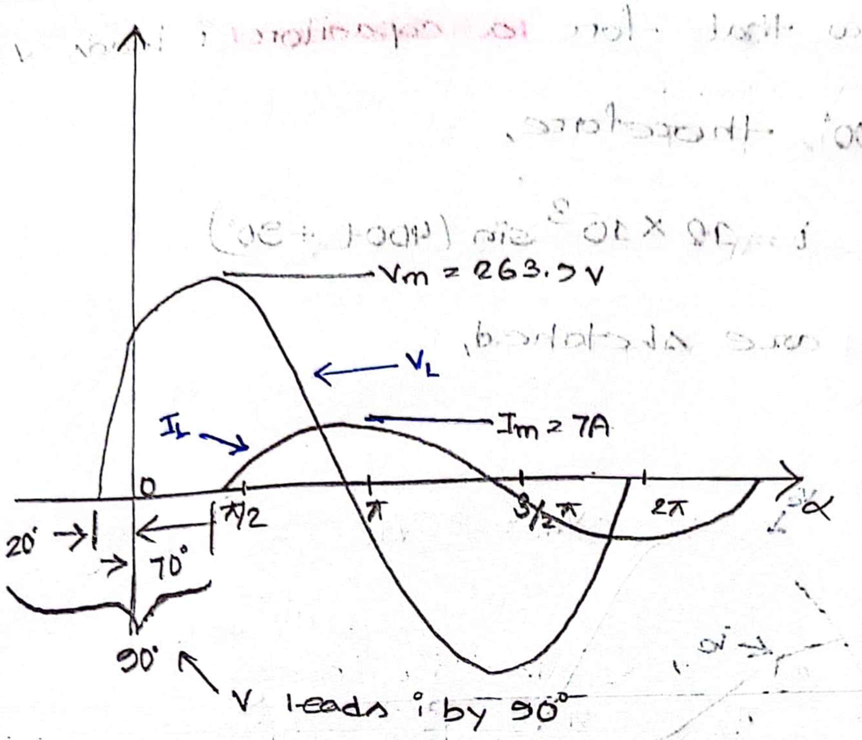
also we know that for a coil/inductor v leads i

by 90°, therefore,

$$v = 263.9 \sin(377t - 70^\circ + 90^\circ)$$

$$\text{or, } v = 263.9 \sin(377t + 20^\circ)$$

The curves are sketched,



Q ✱ Example 14.5 | The voltage across a 1μF capacitor is provided below. What is the sinusoidal expression for the current? Sketch the v and i curves.

$$v = 30 \sin 400t$$

We know, $X_C = \frac{1}{\omega C} = \frac{1}{(400 \text{ rad s}^{-1})(1 \times 10^{-6} F)}$

Capacitor - এর Capacitance $= 2500 \Omega$

Again, $I_m = \frac{V_m}{X_C} = \frac{30}{2500}$

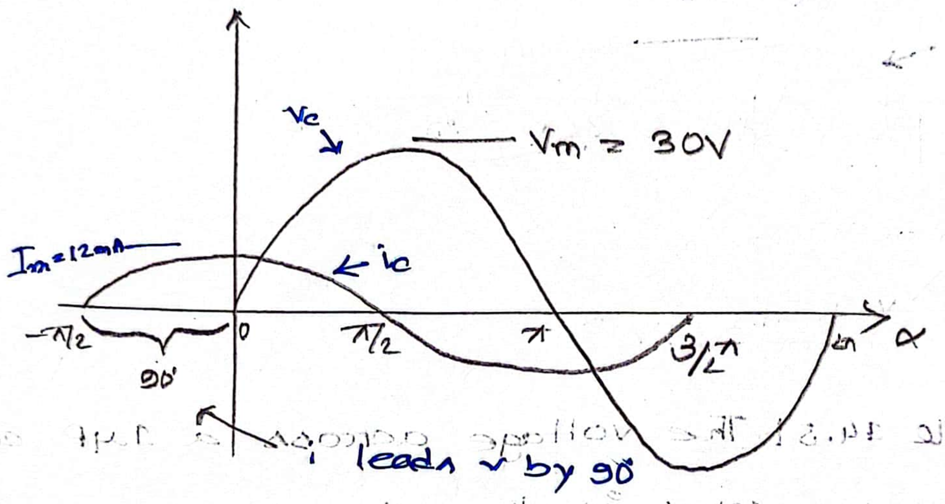
$= 0.0120 A$

$= 12 \text{ mA}.$

and we know that for a capacitor i leads v by 90°, therefore,

$$i = 12 \times 10^{-3} \sin(400t + 90°)$$

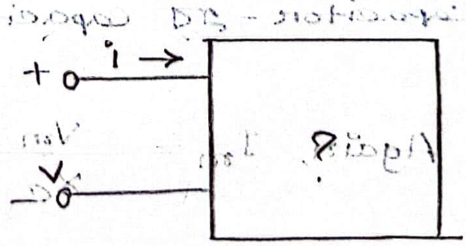
The curves are sketched,



* Example: 14.7 | For the following pairs of voltages and currents, determine whether the element involved is a capacitor, an inductor, or a resistor and determine the value of C, L, or R if sufficient data are provided.

a. $v = 100 \sin(\omega t + 40°)$

$i = 20 \sin(\omega t + 40°)$



⇒ phase difference, $\psi = 40 - 40$

$= 0°$, in-phase

so that, it's a resistor.

Given, $V_m = 100\,V$

$I_m = 20A$

$\therefore R = \frac{V_m}{I_m} = \frac{100}{20}$

$= 5\Omega$

b. $v = 1000 \sin(377t + 10^\circ)$

$i = 5 \sin(377t - 80^\circ)$

$\Rightarrow$ phase difference, $\psi = 10 + 80$

$= 90^\circ$

it's a inductor, cause V leads i by 90°.

Given, $V_m = 1000V$

$I_m = 5A$

$\therefore L = \frac{V_m}{I_m \omega}$

$= \frac{1000}{5 \times 377}$

$= 0.53\,H$

c. $v = 500 \sin(157t + 30^\circ)$

$i = 1 \sin(157t + 120^\circ)$

$\Rightarrow$ phase difference, $\psi = 120 - 30$

$= 90^\circ$

It's a capacitor, cause i leads v by 90°.

Given, $I_m = 1A$

$V_m = 500V$

$\omega = 157$

$$\therefore C = \frac{I_m}{V_m \omega}$$

$$= \frac{1}{500 \times 157}$$

$$= 12.738 \times 10^{-6} F$$

$$= 0.0127 \text{ mF}$$

d. $V = 50 \cos(\omega t + 20°)$

$i = 5 \sin(\omega t + 110°)$

$\Rightarrow i = 5 \cos(\omega t + 110 - 90)$

$= 5 \cos(\omega t + 20°)$

$\therefore$ phase difference, $\varphi = 20 - 20$

$= 0$; in-phase

So that it's a resistor.

Given, $V_m = 50V$

$I_m = 5A$

$$\therefore R = \frac{V_m}{I_m} = \frac{50V}{5A}$$

$$= 10\Omega$$

Phasor Relationships for Circuit Elements

Example: 7. If voltage $v(t) = 6\cos(100t - 30°)$ is applied to a 50μF capacitor, calculate the current, $i(t)$, through the capacitor.

$\Rightarrow$ $v(t) = 6\cos(100t - 30°)$

we know, $X_C = \frac{1}{\omega C} = \frac{1}{(100\ rads^{-1})(50\times 10^{-6})}$

$= 200\ \Omega$

Again, $I_m = \frac{V_m}{X_C}$

$= \frac{6}{200} = 30mA$

And, we know that, for a capacitor i leads v by 90°, therefore,

$i(t) = 30 \cancel{\times 10^{-3}}\ \cancel{\sin}\ \cos(100t - 30° + 90°)$

$= 30 \cancel{\times 10^{-3}} \cos(100t + 60°)\ mA$

Impedance and Admittance

* $Z = \frac{V}{I} = R + jX$

It's Impedance, Unit Ω.

where $R = Re$, Z is the resistance and $X = Im$, Z is the reactance. Positive X is for L and negative X is for C.

* The admittance Y is the reciprocal of impedance, measured in siemens (S).

$$Y = \frac{1}{Z} = \frac{I}{V}$$

Impedances and admittances of passive elements		
Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

(a)

L

$Z = j\omega L$ $\longrightarrow$ Short circuit at dc $\quad \omega = 0;\ Z = 0$

$\omega \to \infty,\ Z \to \infty$

Open circuit at high frequencies

(b)

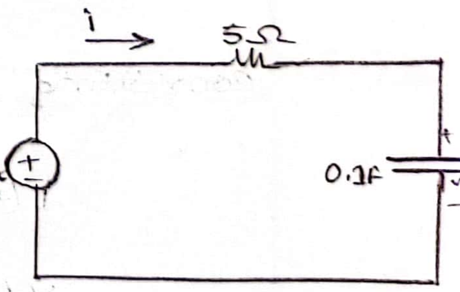
C

$Z = \dfrac{1}{j\omega C}$ $\longrightarrow$ Open circuit at dc $\quad \omega = 0;\ Z \to \infty$

$\omega = \infty,\ Z = 0$

Short circuit at high frequencies

Q * Example: 9.9. Find $v(t)$ and $i(t)$ in the circuit.



$\Rightarrow$ From the voltage source $10\cos 4t$,

$\omega = 4$,

$V_s = 10\angle 0°\ V$

The impedance is

$$Z = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j \times 4 \times 0.1} = 5 - j2.5\,\Omega$$

Hence the current,

$$I = \frac{V_s}{Z} = \frac{10\angle 0^\circ}{5 - j2.5}$$

$$= \frac{10(5+j2.5)}{5^2 + 2.5^2}$$

$$= 1.6 + j0.8$$

$$= 1.789\angle 26.57^\circ A$$

The voltage across the capacitor is,

$$V = IZ_c = \frac{I}{j\omega C} = \frac{1.789\angle 26.57}{j4 \times 0.1}$$

$$= \frac{1.789\angle 26.57}{0.4\angle 90^\circ}$$

$$= 4.47\angle -63.43^\circ V$$

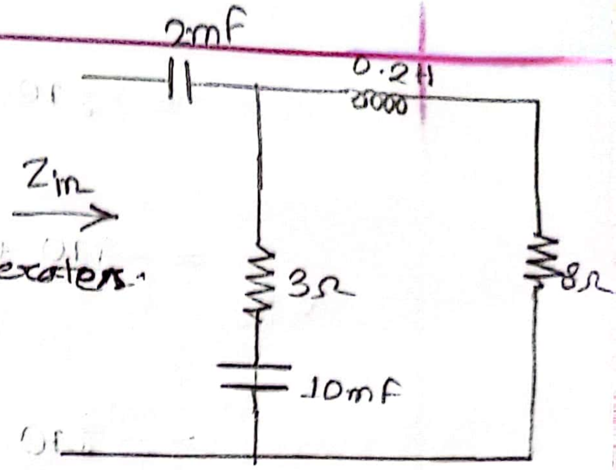
Converting I and V to the time domain, we get,

$$i(t) = 1.789\cos(4t + 26.57^\circ) A$$

$$v(t) = 4.47\cos(4t - 63.43^\circ) V$$

Notice that i(t) leads v(t) by 90° as expected.

Example : 9.10. Find the input impedance of the circuit. Assume that the circuit operates at $\omega = 50$ rads^{-1}.



⇒ Let,

Z_1 = Impedance of the 2mF capacitor

Z_2 = Impedance of the 3Ω resistor in series with the 10 mF capacitor.

Z_3 = Impedance of the 0.2H inductor in series with the 8Ω resistor.

Then

$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j \times 50 \times 2 \times 10^{-3}} = -j10\Omega$$

$$Z_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j \times 50 \times 10 \times 10^{-3}} = (3 - j2)\Omega$$

$$Z_3 = 8 + j\omega L = 8 + j \times 50 \times 0.2$$

$$= (8 + j10)\Omega$$

The input ~~indepan~~ impedance is

$$Z_{in} = Z_1 + Z_2 \| Z_3$$

$$= -j10 + \frac{(3-j2)(8+j10)}{11+j8}$$

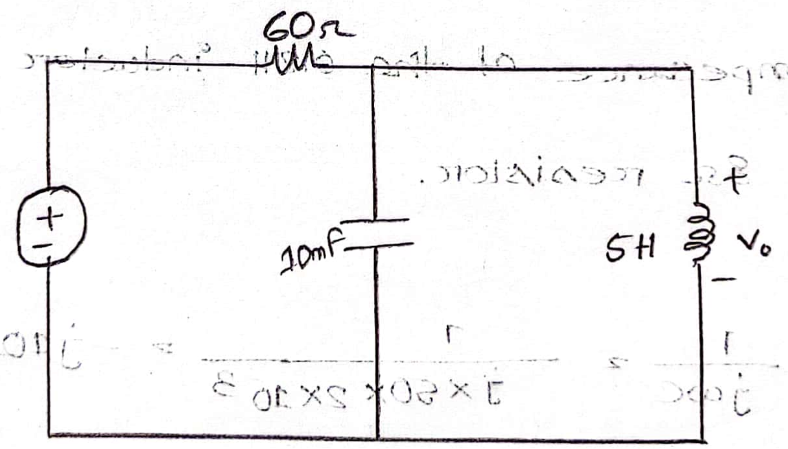
$$= -j10 + \frac{(44+j14)(11-j8)}{11^2+8^2}$$

$$= -j10 + 3.22 - j1.07\,\Omega$$

Thus,

$$Z_{in} = 3.22 - j11.07\,\Omega$$

Q * Example: 9.11 | Determine $v_o(t)$ in the circuit.



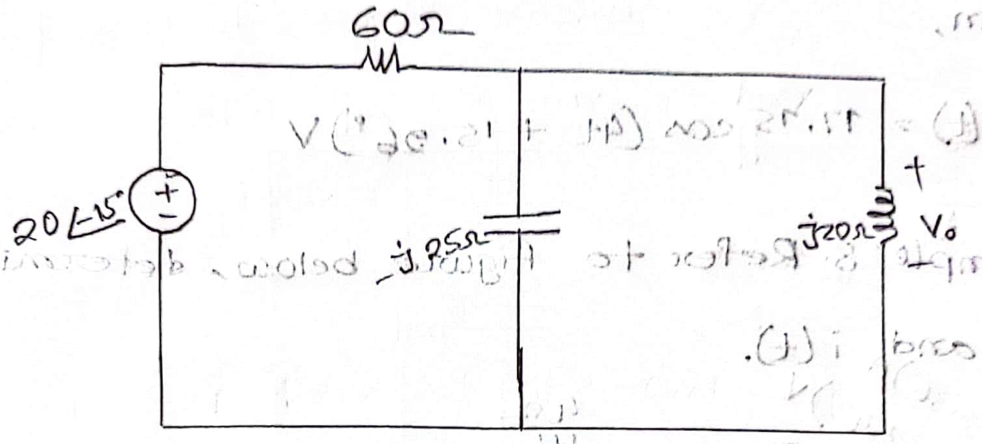
To do the analysis in the frequency domain, we must first transform the time domain circuit to the phasor domain equivalent.

The transform produces,

$$V_s = 20\cos(4t - 15^\circ) \Rightarrow V_s = 20\angle{-15^\circ}\,V, \quad \omega = 4$$

$$10\,mF \Rightarrow \frac{1}{j\omega C} = \frac{1}{j \times 4 \times 10 \times 10^{-3}}$$

$5H \Rightarrow j\omega L = j \times 4 \times 5 = j20\Omega$



Let,

Z_1 = Impedance of the 60-Ω resistor

Z_2 = Impedance of the parallel combination of the 10mF capacitor and the 5H inductor

Then $Z_1 = 60\Omega$ and

$$Z_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20}$$

$$= j100\Omega$$

By the voltage division principle,

$$V_o = \frac{Z_2}{Z_1 + Z_2} V_s = \frac{j100}{60 + j100}(20\angle -15^\circ)$$

$$= (0.8575\angle 30.96^\circ)(20\angle -15^\circ)$$

$$= 17.15\angle 15.96^\circ \text{ V}$$

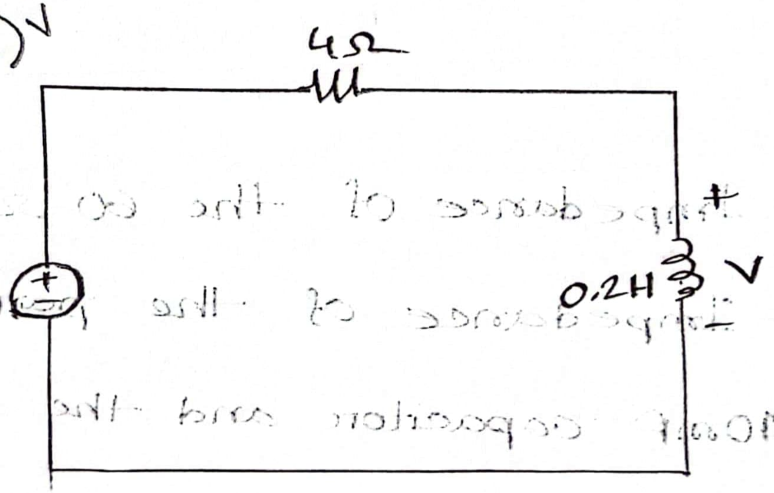
We convert this to the time domain and obtain,

$$V_o(t) = 17.15 \cos(4t + 15.96^\circ)\ V$$

* Example: 8. Refer to figure below, determine $v(t)$ and $i(t)$.

Ans: $i(t) = 1.118 \cos(10t - 26.56^\circ)\ A$

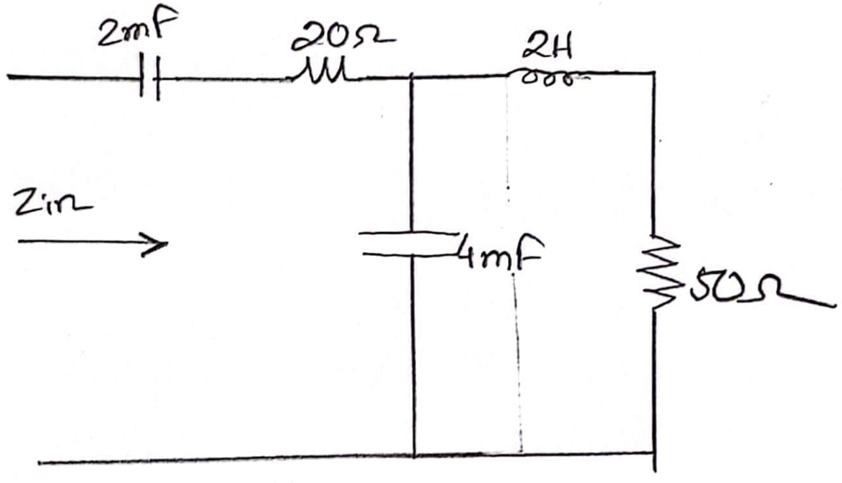
$v(t) = 2.236 \cos(10t + 63.43^\circ)\ V$



$V_s = 5\cos(10t)$

Q. Example: 9. Determine the input impedance of the circuit in figure below at $\omega = 10$ rads^{-1}

Ans: $Z_{in} = 32.38 - j73.76$



Let,

Z_1 = Impedance of the 2-mF capacitor

Z_2 =